

APEXPLANET SOFTWARE PVT. LTD.

Cybersecurity & Ethical Hacking Internship

TASK 1

Foundations of Cybersecurity — Lab Report

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Offer Letter ID: APSPL2641722

Internship Duration: 11 July 2026 – 08 September 2026

Domain: Cybersecurity & Ethical Hacking

Internship Snapshot

Company	ApexPlanet Software Pvt. Ltd.
Domain	Cybersecurity & Ethical Hacking
Offer Letter ID	APSPL2641722
Duration	11 July 2026 – 08 September 2026 (60 Days)
Stipend	Unpaid
Current Task	Task 1 of 5 — Foundations of Cybersecurity (Days 1–12)

1. Objective

The objective of Task 1 is to establish a self-contained cybersecurity lab and build foundational competency in core security concepts, Linux systems, networking, and cryptography. This includes deploying a Kali Linux attacker machine and a Metasploitable2 target machine on an isolated Host-Only network, and using industry-standard tools — Nmap, Wireshark, Burp Suite, Netcat, and OpenSSL — to observe and analyze network and application traffic in a safe, controlled environment.

2. Lab Environment Setup

2.1 Virtualization Platform

The lab was built using Oracle VirtualBox, hosting two virtual machines configured to communicate over a private network segment.

Component	Role	Details
Kali Linux	Attacker / Analysis machine	Pre-loaded with Nmap, Wireshark, Burp Suite, Metasploit, Netcat, OpenSSL
Metasploitable2	Target machine	Intentionally vulnerable Linux VM used for safe practice
Host-Only Network	Adapter 1	Isolated network for VM-to-VM traffic (used for testing)
NAT Network	Adapter 2	Provides internet access to the Kali VM for updates

2.2 Network Addressing

Machine	Interface	IP Address
Kali Linux	eth0 (Host-Only)	192.168.56.102
Metasploitable2	eth0 (Host-Only)	192.168.56.101

3. Connectivity Verification — Ping Test

Basic ICMP connectivity between the two virtual machines was verified before proceeding to deeper scanning and analysis.

```
$ ping 192.168.56.101
PING 192.168.56.101: 56 data bytes
64 bytes from 192.168.56.101: icmp_seq=0 ttl=64 time=0.412 ms
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.389 ms
```

Result: Both machines responded successfully, confirming the Host-Only network was correctly configured and the target was reachable.

4. Reconnaissance — Nmap Scan

An Nmap scan was performed against the Metasploitable2 target to identify open ports and running services, which informed the subsequent Wireshark and Burp Suite exercises.

```
$ nmap -sV 192.168.56.101
```

```

Session Actions Edit View Help
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output
MISC:
  -G: Enable IPv6 scanning
  -A: Enable OS detection, version detection, script scanning, and traceroute
  --datadir <dirname>: Specify custom Nmap data file location
  --send-eth/--send-ip: Send using raw ethernet frames or IP packets
  --privileged: Assume that the user is fully privileged
  --unprivileged: Assume the user lacks raw socket privileges
  -v: Print version number
  -h: Print this help summary page.
EXAMPLES:
  nmap -v -A scanme.nmap.org
  nmap -v -sn 192.168.0.0/16 10.0.0.0/8
  nmap -v -iR 100000 -Pn -p 80
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES

[sai@kali:~]$ nmap -sV 192.168.56.101
Starting Nmap 7.90 ( https://nmap.org ) at 2026-07-08 13:22 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0042s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 5ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          netkit-rshd
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi       GNU Classpath gmireregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp           ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-JubuntuS
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8000/tcp  open  ajp13          Apache jkerv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:58:CF:13 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.93 seconds

[sai@kali:~]$

```

5. Network Traffic Analysis — Wireshark

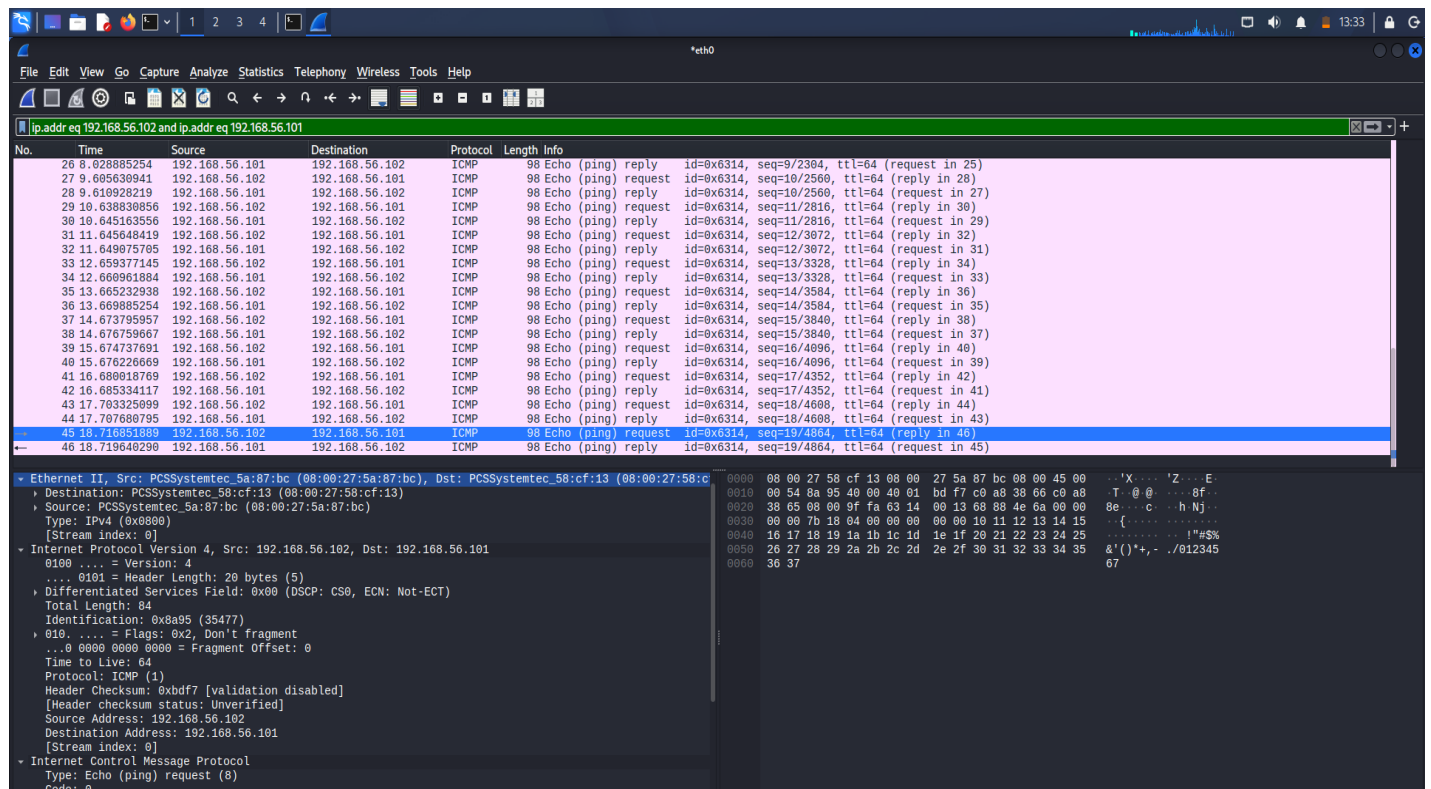
Wireshark was used to capture and inspect live traffic exchanged between Kali Linux and Metasploitable2 on the eth0 (Host-Only) interface.

5.1 Procedure

- Launched Wireshark and selected the eth0 interface (192.168.56.102, Host-Only).
- Started a live capture, then generated ICMP traffic using ping 192.168.56.101.
- Stopped the capture and applied the display filter: icmp
- Inspected an Echo Request packet, expanding the Ethernet II, IPv4, and ICMP layers.
- Saved the capture as task1_ping_capture.pcapng.

5.2 Observation

The capture confirmed successful ICMP Echo Request / Echo Reply exchanges between 192.168.56.102 and 192.168.56.101, validating end-to-end connectivity at the network layer and demonstrating packet-level inspection using a protocol analyzer.

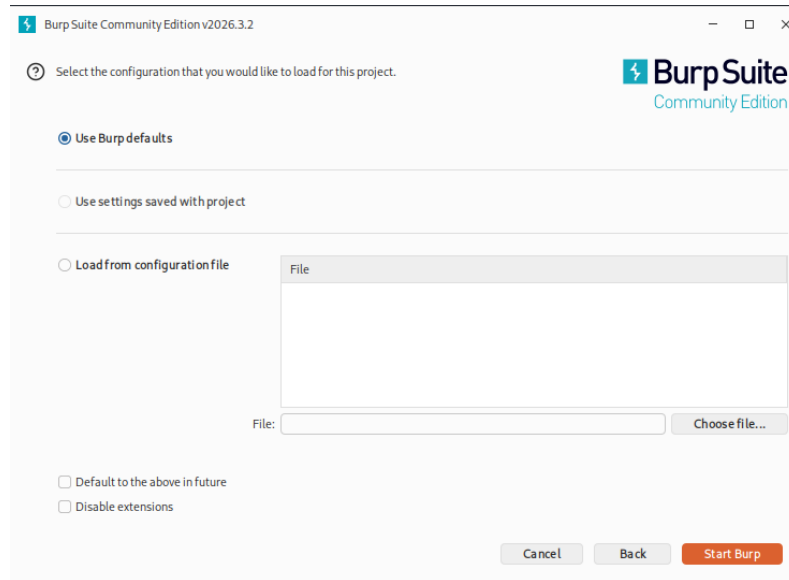


6. Web Traffic Interception — Burp Suite

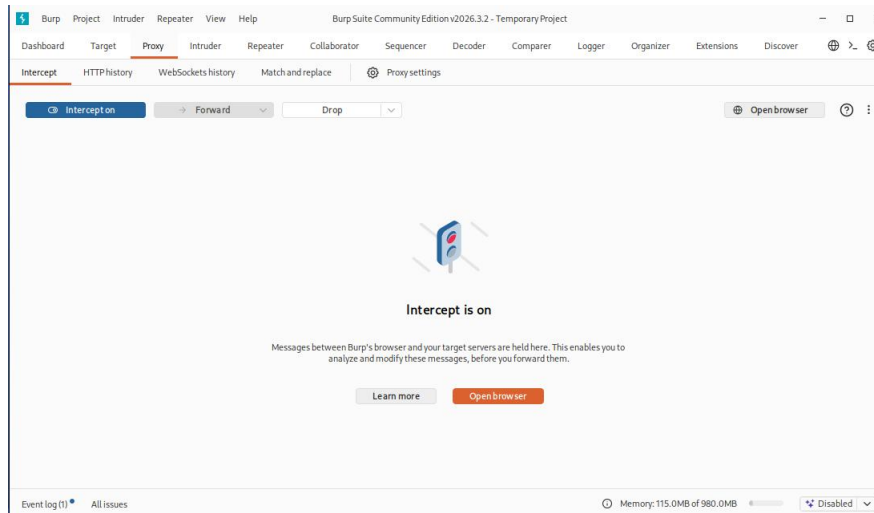
Burp Suite's intercepting proxy was used to capture and forward an HTTP request from Firefox to the Metasploitable2 web server.

6.1 Procedure

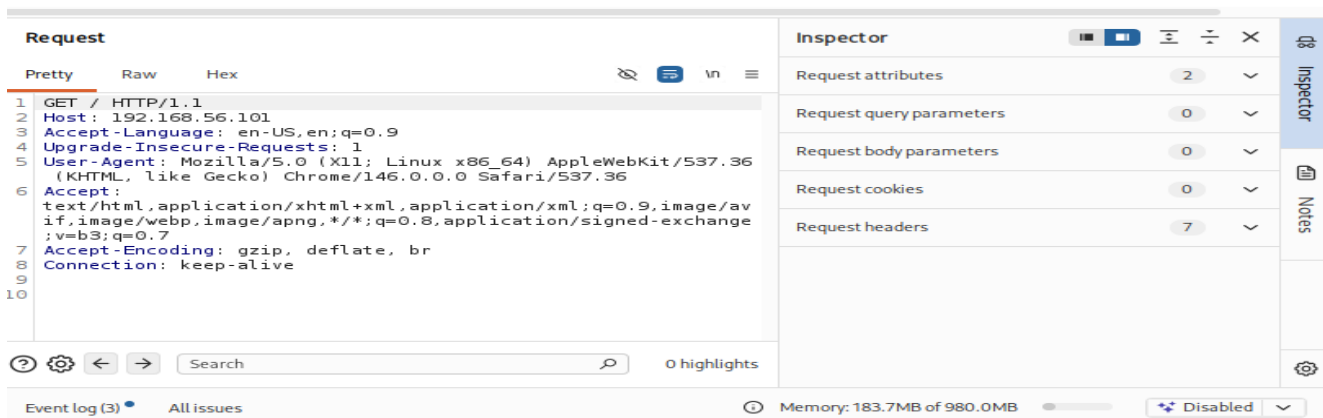
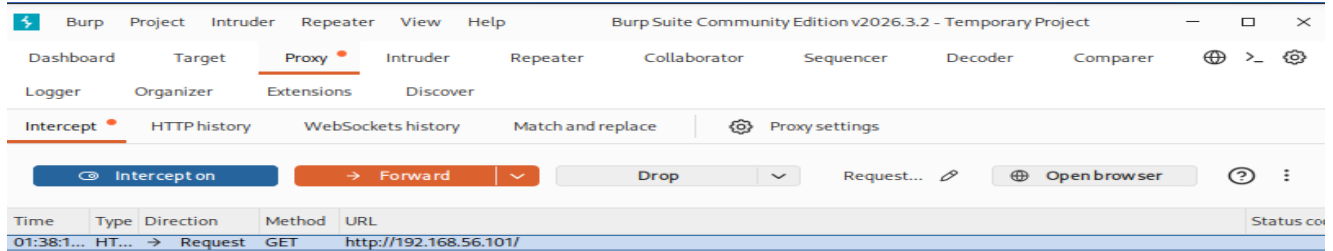
- Started Burp Suite with a Temporary Project using Burp Defaults.



- Enabled Intercept under Proxy → Intercept.



- Browsed to <http://192.168.56.101> in Firefox (configured to route through the Burp proxy).
- Reviewed the intercepted raw HTTP GET request before forwarding it.

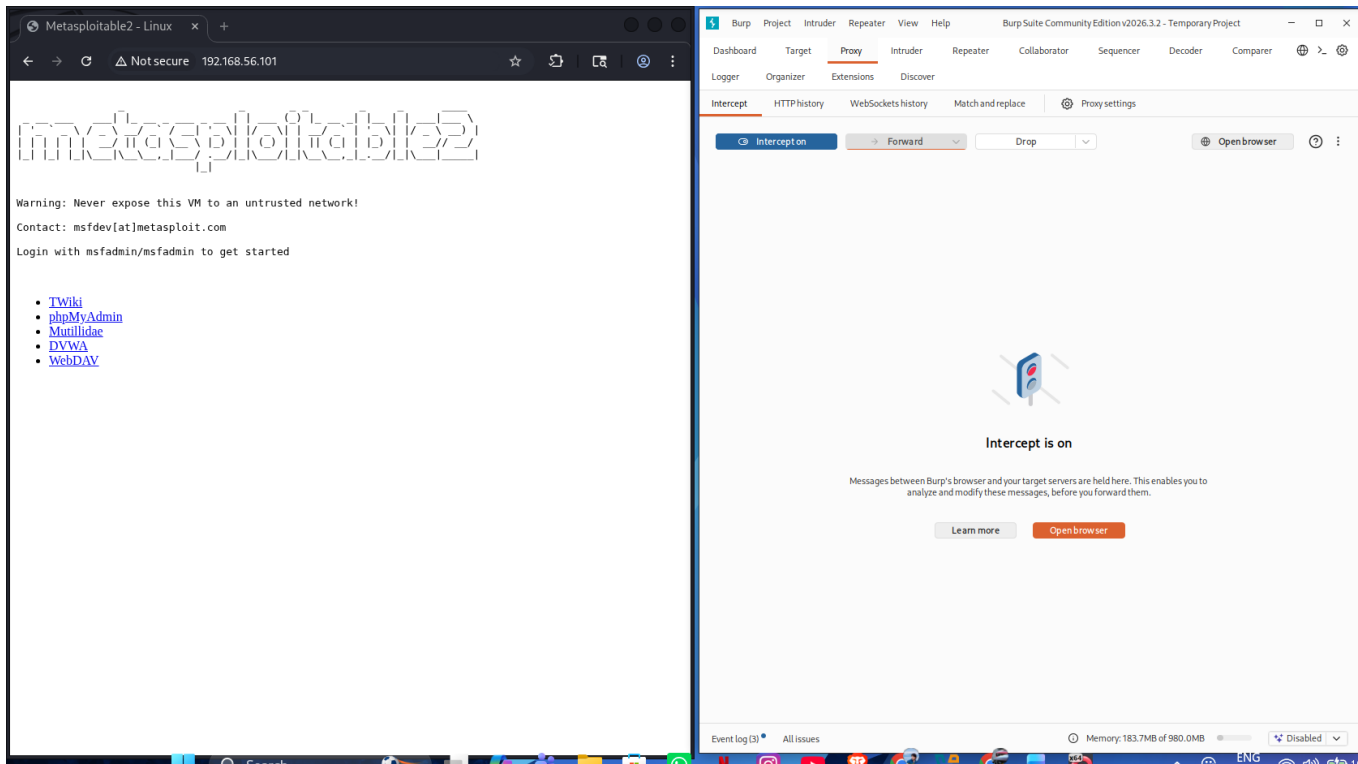


- Forwarded the request and confirmed the Metasploitable2 home page loaded successfully.

```
GET / HTTP/1.1
Host: 192.168.56.101
User-Agent: Mozilla/5.0
```

6.2 Observation

This demonstrated how an intercepting proxy allows a security analyst to pause, inspect, and modify web requests in transit — a core technique used later in web application security testing.



7. Linux Fundamentals

Core Linux commands were exercised to demonstrate navigation, file management, permissions, and system/network information gathering.

Command	Purpose
pwd	Print current working directory
ls	List files and directories
mkdir Task1	Create a new directory
touch demo.txt	Create an empty file
echo "Hello ApexPlanet" > demo.txt	Write text into a file
cat demo.txt	Display file contents
cp demo.txt demo2.txt	Copy a file
mv demo2.txt notes.txt	Rename / move a file
chmod 755 demo.txt	Change file permissions
whoami	Display the current logged-in user
hostname	Display the machine hostname
ip addr	Display network interfaces and IP addresses

```

(sai@kali)~$ ls
Desktop  Documents  Downloads  Music  Pictures  Projects  Public  Templates  Videos

(sai@kali)~$ cd

(sai@kali)~$ mkdir Task1

(sai@kali)~$ cd Task1

(sai@kali)~/Task1$ touch demo.txt

(sai@kali)~/Task1$ echo "Hello ApexPlanet" > demo.txt

(sai@kali)~/Task1$ cat demo.txt
Hello ApexPlanet

(sai@kali)~/Task1$ cp demo.txt demo2.txt

(sai@kali)~/Task1$ mv demo2.txt notes.txt

(sai@kali)~/Task1$ chmod 755 demo.txt

(sai@kali)~/Task1$ whoami
sai

(sai@kali)~/Task1$ hostname
kali

(sai@kali)~/Task1$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:b7:bc brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 569sec preferred_lft 569sec
    inet6 fe80::50f9:6707:b27c:3fe9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000

```

8. Cryptography — OpenSSL

Symmetric file encryption and decryption were demonstrated using OpenSSL's AES-256-CBC cipher.

```

$ echo "Hello ApexPlanet" > message.txt

# Encrypt
$ openssl enc -aes-256-cbc -salt -in message.txt -out message.enc

# Decrypt
$ openssl enc -aes-256-cbc -d -in message.enc -out decrypted.txt

# Verify
$ cat decrypted.txt
Hello ApexPlanet

```

Result: The decrypted output matched the original plaintext exactly, confirming correct encryption and decryption using AES-256-CBC.

```
Session Actions Edit View Help
sai@kali: ~
$ openssl version
OpenSSL 3.0.2 7 Apr 2026 (Library: OpenSSL 3.0.2 7 Apr 2026)
$ mkdir CryptoDemo
$ pwd
/home/sai
$ echo "Hello ApexPlanet" > message.txt
$ cat message.txt
Hello ApexPlanet
$ openssl enc -aes-256-cbc -salt -in message.txt -out message.enc
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
$ ls
CryptoDemo Desktop Documents Downloads message.enc message.txt Music Pictures Projects Public Task1 Templates Videos
$ cat message.enc
Salted__Rp**3eq5zQ**6e# *c*I c!j2*****
$ openssl enc -aes-256-cbc -d -in message.enc -out decrypted.txt
enter AES-256-CBC decryption password:
** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
$ cat decrypted.txt
Hello ApexPlanet
$ ls -l
total 56
drwxrwxr-x 2 sai sai 4096 Jul 9 01:50 CryptoDemo
-rw-rw-r-- 1 sai sai 17 Jul 9 01:53 decrypted.txt
drwxr-xr-x 2 sai sai 4096 Jul 8 10:03 Desktop
drwxr-xr-x 2 sai sai 4096 Jul 8 10:03 Documents
drwxr-xr-x 2 sai sai 4096 Jul 8 10:03 Downloads
-rw-rw-r-- 1 sai sai 48 Jul 9 01:51 message.enc
-rw-rw-r-- 1 sai sai 17 Jul 9 01:51 message.txt
```

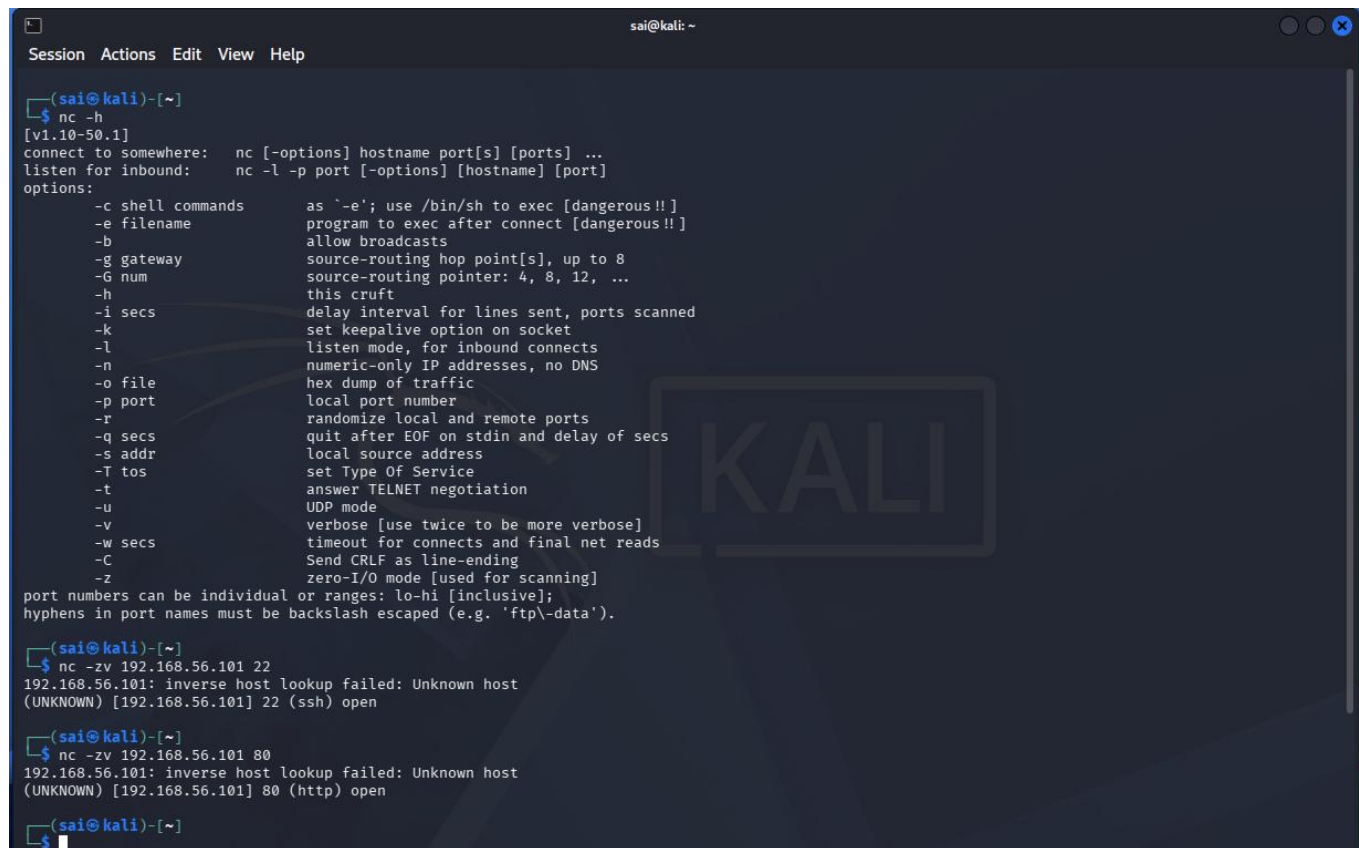
9. Tool Verification — Netcat

Netcat's presence and usability on Kali Linux was confirmed by displaying its help output.

```
$ nc -h

# Metasploitable2 machine has SSH (port 22) open, can test connectivity
$ nc -zv 192.168.56.101 22

# Now test another known open port, such as HTTP
$ nc -zv 192.168.56.101 80
```



```
sai@kali: ~  
Session Actions Edit View Help  
  
(sai@kali)-[~]  
$ nc -h  
[v1.10-50.1]  
connect to somewhere: nc [-options] hostname port[s] [ports] ...  
listen for inbound: nc -l -p port [-options] [hostname] [port]  
options:  
-c shell commands      as '-e'; use /bin/sh to exec [dangerous!!]  
-e filename            program to exec after connect [dangerous!!]  
-b                    allow broadcasts  
-g gateway            source-routing hop point[s], up to 8  
-G num                source-routing pointer: 4, 8, 12, ...  
-h                    this cruft  
-i secs              delay interval for lines sent, ports scanned  
-k                    set keepalive option on socket  
-l                    listen mode, for inbound connects  
-n                    numeric-only IP addresses, no DNS  
-o file              hex dump of traffic  
-p port              local port number  
-r                    randomize local and remote ports  
-q secs              quit after EOF on stdin and delay of secs  
-s addr              local source address  
-T tos               set Type Of Service  
-t                    answer TELNET negotiation  
-u                    UDP mode  
-v                    verbose [use twice to be more verbose]  
-w secs              timeout for connects and final net reads  
-C                    Send CRLF as line-ending  
-z                    zero-I/O mode [used for scanning]  
  
port numbers can be individual or ranges: lo-hi [inclusive];  
hyphens in port names must be backslash escaped (e.g. 'ftp-data').  
  
(sai@kali)-[~]  
$ nc -zv 192.168.56.101 22  
192.168.56.101: inverse host lookup failed: Unknown host  
(UNKNOWN) [192.168.56.101] 22 (ssh) open  
  
(sai@kali)-[~]  
$ nc -zv 192.168.56.101 80  
192.168.56.101: inverse host lookup failed: Unknown host  
(UNKNOWN) [192.168.56.101] 80 (http) open  
  
(sai@kali)-[~]  
$
```

10. Conclusion

Task 1 successfully established the cybersecurity lab environment required for the remainder of the internship. A working Kali Linux and Metasploitable2 setup was configured on an isolated Host-Only network, connectivity was verified, and hands-on practice was completed across reconnaissance (Nmap), packet analysis (Wireshark), web traffic interception (Burp Suite), Linux system administration, and applied cryptography (OpenSSL). This foundation directly supports Task 2 (Network Security & Scanning) and the subsequent tasks in the internship roadmap.

Deliverables Submitted

- Lab Setup Report (this document)
- GitHub Repository: Cybersecurity-Task1 (README, Linux-Commands.md, Networking.md, Cybersecurity-Basics.md, Cryptography.md, Screenshots/, Reports/, task1_ping_capture.pcapng)
- 5-minute Demo Video (uploaded to LinkedIn Featured section)